

Amin Shahpur

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EDUCATION

George Mason University

Bachelor of Science in Computer Science, GPA 3.7

Aug. 2022 – Dec. 2024

Fairfax, VA

Northern Virginia Community College

Associate of Science in Computer Science, GPA 4.0

Aug. 2019 – Aug. 2021

Sterling, VA

- Relevant Courses: Data Structures & Algorithms, Object-Oriented Programming (Java), Databases (SQL), Low-Level Programming (C), Analysis of Algorithms

EXPERIENCE

Lead Role Player Contractor

Vertical Jobs, Inc.

Sep. 2022 – Present

Various Locations, Eastern United States

- Supported federal and military agencies (Department of State, FBI, USPIS, U.S. Army, USMC) by simulating high-risk training environments for tactical readiness and to train military personnel and law enforcement agencies in crisis response and scenario-based operations.
- Participated in counterterrorism exercises, vehicle pursuits, and carjacking scenarios to assess and enhance preparedness.
- Portrayed roles such as victims, witnesses, and distressed individuals in mock trials and training exercises.
- Ensured safety and maintained confidentiality during exercises.
- Provided feedback on training exercises to enhance methodologies and realism.

Operations Team Lead

Xator Corporation (acquired by Parsons)

Aug. 2021 – Feb. 2022

Dulles, VA

- Worked on the Operation Allies directive for the Department of Homeland Security to aid the resettlement and assistance of refugees.
- Kept supervisor informed of needs and potential problems in assigned areas.
- Maintained confidentiality and privacy standards vital to the safety and success of the project.

Peer Mentor

College Steps

Dec 2020 – May 2021

Sterling, VA

- Provided one-on-one support and guidance to students with disabilities, fostering a welcoming and inclusive environment.
- Acted as a reliable and empathetic resource for mentees, helping them navigate academic and personal challenges.
- Collaborated with students, faculty, and staff to advocate for accessible resources and accommodations for mentees.
- Collaborated with fellow mentors and program coordinators to share insights, coordinate events, and improve the overall mentorship program.

PROJECTS

Unix Buffer Manager

Apr. 2024

- Developed a Unix-based buffer management system using C, enabling users to create, modify, and manage multiple text buffers interactively within a shell interface.
- Implemented process management and inter-process communication and pipes to allow concurrent execution of commands, background processing, and controlled process termination.

- Optimized system logging and error handling by integrating structured logging functions to track process states, buffer activity, and user commands, ensuring efficient debugging and maintainability.

MineSweeper

Sep. 2023

- Designed and developed the classic MineSweeper game.
- Utilized fundamental data structures to implement a dynamic array system for creating and managing the game board, enhancing user experience and performance.
- Leveraged object-oriented principles to organize and maintain codebase, ensuring scalability and modularity.

Polyglot Discord Bot

May 2024

- Developed a Discord bot for real-time message translation, utilizing the Googletrans library to enable seamless multilingual communication in Discord servers.
- Implemented an interactive command system with features like automatic translation sessions and random GIF generation based on user messages.
- Integrated API handling and chatbot functionalities, optimizing message parsing and response efficiency for a dynamic user experience.

Blockchain Tree Implementation

Nov. 2023

- Designed and implemented a blockchain data structure in Java, constructing linked blocks that securely store financial transactions using Merkle trees and cryptographic hashing (SHA-256).
- Developed a priority queue-based transaction system, ensuring optimal processing order using a custom singly linked list to prioritize transactions with higher fees.
- Optimized transaction verification with $O(\log N)$ complexity, leveraging Merkle tree hashing for efficient blockchain integrity checks without full-chain traversal.

TECHNICAL SKILLS

Languages: Java, Python, C, SQL, JavaScript, HTML, CSS, MATLAB

Developer Tools: Linux, Git, Google Cloud Platform, VS Code, Eclipse, OpenAI

Certifications: LPI Linux Essentials, AWS Certified Cloud Practitioner (In Progress)